Co-occurrence of Beckwith–Wiedemann Syndrome and early-onset colorectal cancer

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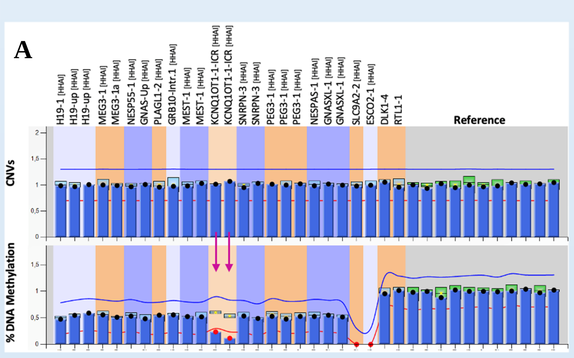
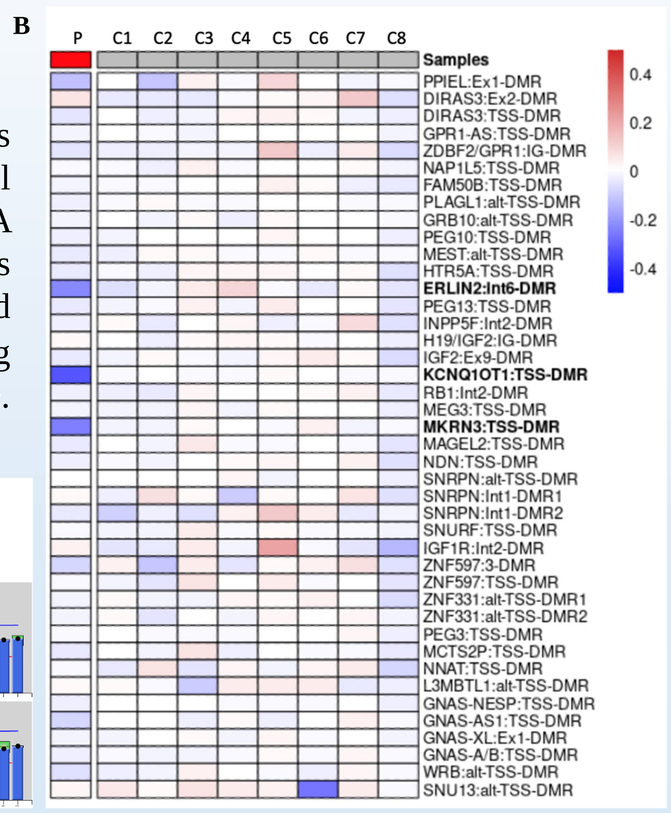
INTRODUCTION

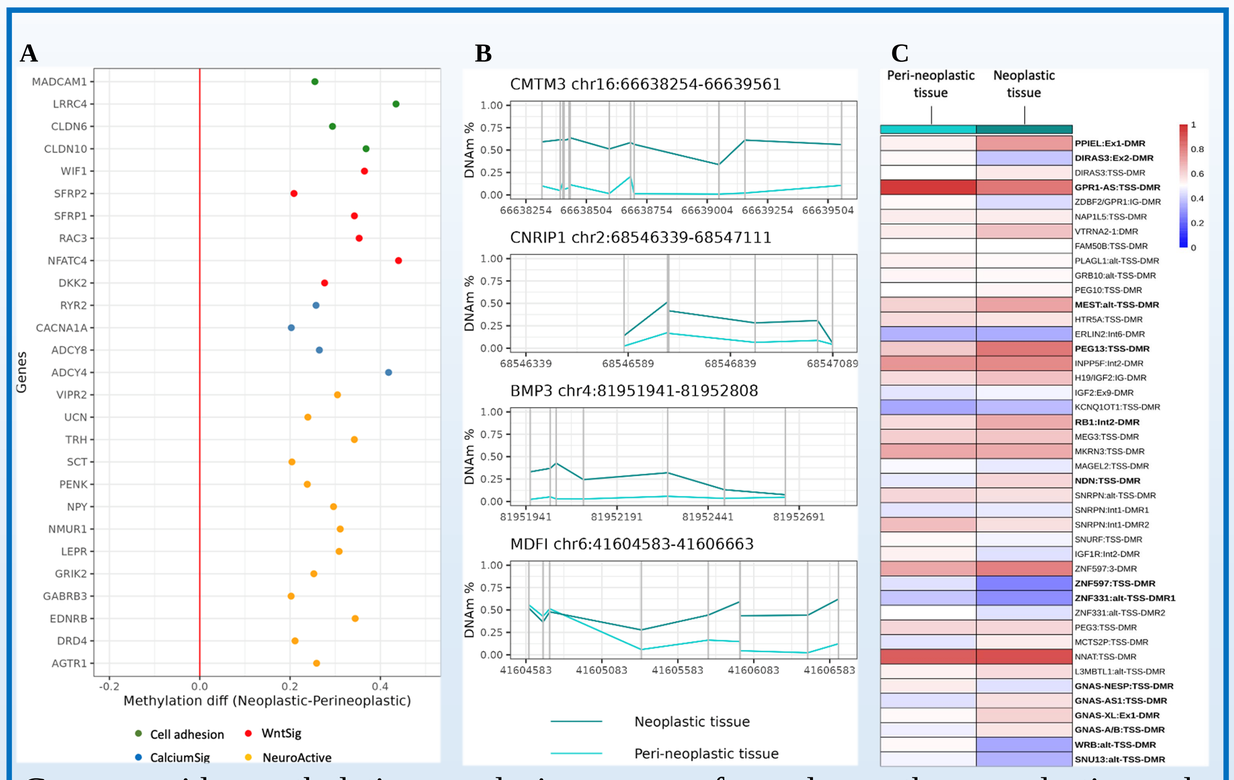
BWSp is a disorder predisposing to tumors of embryonic origin that arise during childhood. Data on tumor risk and histotype prevalence in adult BWSp patients are very limited because the clinical phenotype mitigates with age and follow-up of large series of adults with BWSp has not been performed. However, several studies report cases of co-occurrence of BWSp and endocrine tumors or others in early adulthood, suggesting that the cancer risk in BWSp could be relevant also later than childhood. Colorectal cancer (CRC) is an adult-onset carcinoma representing the third most common cancer and the second leading cause of cancer related deaths in the world. Early-onset (<45 years of age) CRC (EO-CRC) accounts for 5% of the CRC cases and is associated with cancer predisposing genetic factors in half of them. In this study, we describe the case of a woman affected by BWSp, who developed EO-CRC at age 27. Suspected clinical diagnosis of BWSp was based on the presence of ear pits and heterometry of the upper and lower limbs (clinical score for BWSp = 3), then confirmed by the finding of IC2 LoM. Genome-wide epigenetic and genetic analyses were performed to look for a possible link between BWSp and EO-CRC.

RESULTS

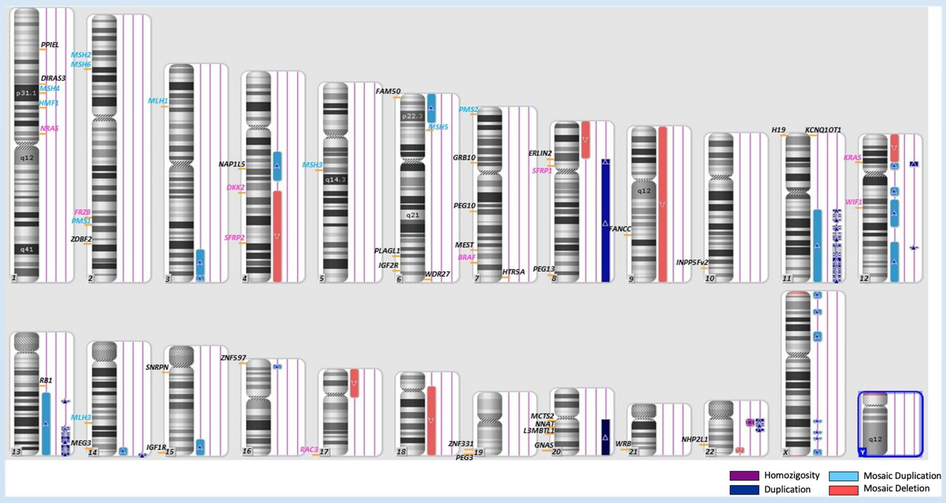
Methylation analysis of the iDMRs was performed on peripheral blood leukocytes (PBL) DNA of the proband and control subjects by multiple MS-MLPA (A) and methylation array (B), revealing IC2 LoM and a mild MLID profile.





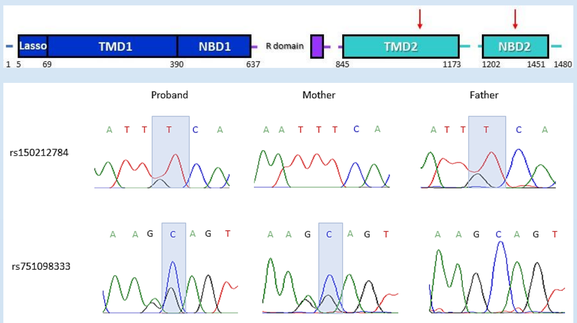


Genome-wide methylation analysis was performed on the neoplastic and peri-neoplastic colon tissues. Results revealed a general epigenome instability of the neoplastic tissue, showing methylation changes that affected genes of the WNT signaling pathway(A), CRC biomarkers (B) and imprinted loci (C).



Karyoview of the neoplastic tissue analyzed by SNP-Array. The Results demonstrate many copy number changes, both duplications and deletions affecting most chromosomes.

Variant ID	Nucleotide change	Aminoacidic change	Allele frequency (gnomAD)	PolyPhen-2	ACMG
rs150212784	c.3154T>G	p.Phe1052Val	0.0006282	possibly damaging	likely pathogenic
rs751098333	c.4054C>G	p.Gln1352Glu	0.00001415	possibly damaging	likely pathogenic



To investigate the presence of germline genetic variants predisposing to CRC, whole-exome sequencing of PBL DNA of the patient was performed. Two very rare SNVs of the cystic fibrosis transmembrane conductance regulator (*CFTR*) gene were found in heterozygosity.

CONCLUSIONS

We found IC2 LoM as cause of BWSp, pathogenic *CFTR* variants as possible germline lesions predisposing to CRC, and a chromosomal and epigenome instability as altered molecular profile of the tumor genome. The development of a very EO-CRC can be explained by the adverse circumstance of co-occurrence of *CFTR* mutations and IC2 LoM, likely responsible for cancer initiation, further triggered by somatic events enhancing the WNT/beta-catenin pathway. By the clinical point of view, our study supports the need of extending cancer surveillance beyond the interval normally recommended for BWSp patients.

References

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